



Acoustic Test Report

APPLICANT : Magne AI Global tech limited
EQUIPMENT : MAG1
BRAND NAME : MaQ
MODEL NAME : MA1
STANDARD : EN50332-2:2013
TEST DATE(S) : Mar. 16, 2026

The maximum sound pressure level of equipment under test complies with, and the maximum output voltage complies with EN 50332-2.

The test results in this report apply exclusively to the tested model / sample. Without written approval of Sporton International Inc. (KunShan), the test report shall not be reproduced except in full.

Jason Jia

Approved by: Jason Jia



Sporton International Inc. (Kunshan)

**No. 1098, Pengxi North Road, Kunshan Economic Development Zone Jiangsu Province 215300
People's Republic of China**



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REVISION HISTORY

REPORT NO.	VERSION	DESCRIPTION	ISSUED DATE
EO612311	Rev. 01	Initial issue	May 25, 2026

Conformity Assessment Condition:

1. The test results (PASS/FAIL) with all measurement uncertainty excluded are presented against the regulation limits or in accordance with the requirements stipulated by the applicant/manufacture who shall bear all the risks of non-compliance that may potentially occur if measurement uncertainty is taken into account.
2. The measurement uncertainty please refer to each test result in the section "Measurement Uncertainty"

Disclaimer:

The product specifications of the EUT presented in the test report that may affect the test assessments are declared by the manufacturer who shall take full responsibility for the authenticity.



1. General Description of Equipment under Test

1.1 Applicant

Magne AI Global tech limited

FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

1.2 Manufacturer

Magne AI Global tech limited

FLAT 1019B,10/F,LIVEN HOUSE,NO.61-63 KING YIP STREET KWUN TONG HK

1.3 Product Feature of Equipment Under Test

Product Feature	
Equipment	MAG1
Brand Name	MaQ
Model Name	MA1
IMEI Code	016813000002475/016813000002483

Remark:

1. The above EUT's information was declared by manufacturer. Please refer to the specifications or user's manual for more detailed description.
2. The device is not sold with earphone thus only EN50332-2 is assessed.

1.4 Modification of EUT

No modifications are made to the EUT during all test items.



1.5 Testing Facility

Sporton International Inc. (Kunshan) is accredited to ISO/IEC 17025:2017 by American Association for Laboratory Accreditation with Certificate Number 5145.02.

Test Lab.	Sporton International Inc. (Kunshan)
Test Site Location	No. 1098, Pengxi North Road, Kunshan Economic Development Zone Jiangsu Province 215300 People's Republic of China TEL : +86-512-57900158
Test Site No.	Site No. : AC01-KS Test Engineer : Jarvis

1.6 Test Software

Item	Site	Manufacturer	Name	Version
1.	AC01-KS	R&S	1GA48	2.0.0
2.	AC01-KS	B&K	BZ-5829-N	25.1.0.198

1.7 Supported Unit Used in Test

Item	Equipment	Trade Name	Model Name	Data Cable	Power Cord
1.	Type C to Audio cable	UGREEN	AV161	Unshielded 12.5cm	N/A

1.8 Applied Standards

According to the specifications of the manufacturer, the EUT must comply with the requirements of the following standard:

1. The measurement was implemented in accordance with EN 50332-2.
2. Test requirement and limitation in accordance with clause 10 of IEC 62368-1

Note: All test items were verified and recorded according to the standards and without any deviation during the test.

2. Measurement Method

This test method is based on the use of a Head and Torso Simulator (HATS), B&K type 4128 equipped with ear simulators type 4158 and 4159 in accordance with IEC 60318-7.

Test source signal used to measure the maximum sound pressure level is a program simulation noise, as defined in HD 483.1 S2., and its crest factor range is between 1.8 ~ 2.2. It is fed into the system simulator over the air to the device under test.

The sound pressure level measured by the ear simulator microphone represents the pressure found at eardrum level, and that levels have been correlated by the acoustic free-field transfer function of the simulator. The results are given as free field related A-weighted and in dB unit.

Test is repeated five times for each ear, and the earphones are removed and repositioned each test prior to each measurement. The equipment under test is set in maximum volume value.

The maximum sound pressure level considered as the test result is the mean value of all L_{Aeq} measurements.

For headset/earphone, the simulated programme signal characteristic voltage of analogue headphone input is the input signal voltage when the sound pressure level reaches 94 dB SPL A-weighted, and test procedure (same procedure above for player) is repeated five times.

For the maximum output voltage measurement, the V_m shall be defined as unweighted true r.m.s. voltage at the load, using an averaging time of 30 s or more. Player output shall be loaded with a resistive load of 32 Ω .

For digital listening devices, the test signal shall be applied to the listening device with an r.m.s. amplitude of - 10 dBFS, where 0 dBFS is defined as being the maximum RMS amplitude of a sinusoidal signal corresponding to the full scale of the digital interface.



3. Test Equipment

Manufacturer	Name of Equipment	Type/Model	Calibration Date	Due Date
B&K	HATS	4128-C-001	NCR	NCR
B&K	Audio Power Amplifier	2716-C-001	Calibrated as a whole system	N/A
B&K	Conditioning Amplifier	2690-OS2	Calibrated as a whole system	N/A
R&S	Audio Analyzer	UPV	Oct. 10, 2025	Oct. 09, 2026
B&K	Sound calibrator	4231	Jul. 10, 2025	Jul. 09, 2026
B&K	LAN-XI	3160-A-042	Jan. 05, 2026	Jan. 04, 2027

The Calibrated as a whole system is calibrated by the acoustical calibrator, 4231, and considers cable connections for accuracy, before starting test.

4. Test Result

4.1 Test Environment and Test Engineer

Temperature	Relative Humidity	Atmospheric Pressure	Test Engineer
21~23℃	41~42%	1017hPa	Jarvis

4.2 Test Results for Music Mode

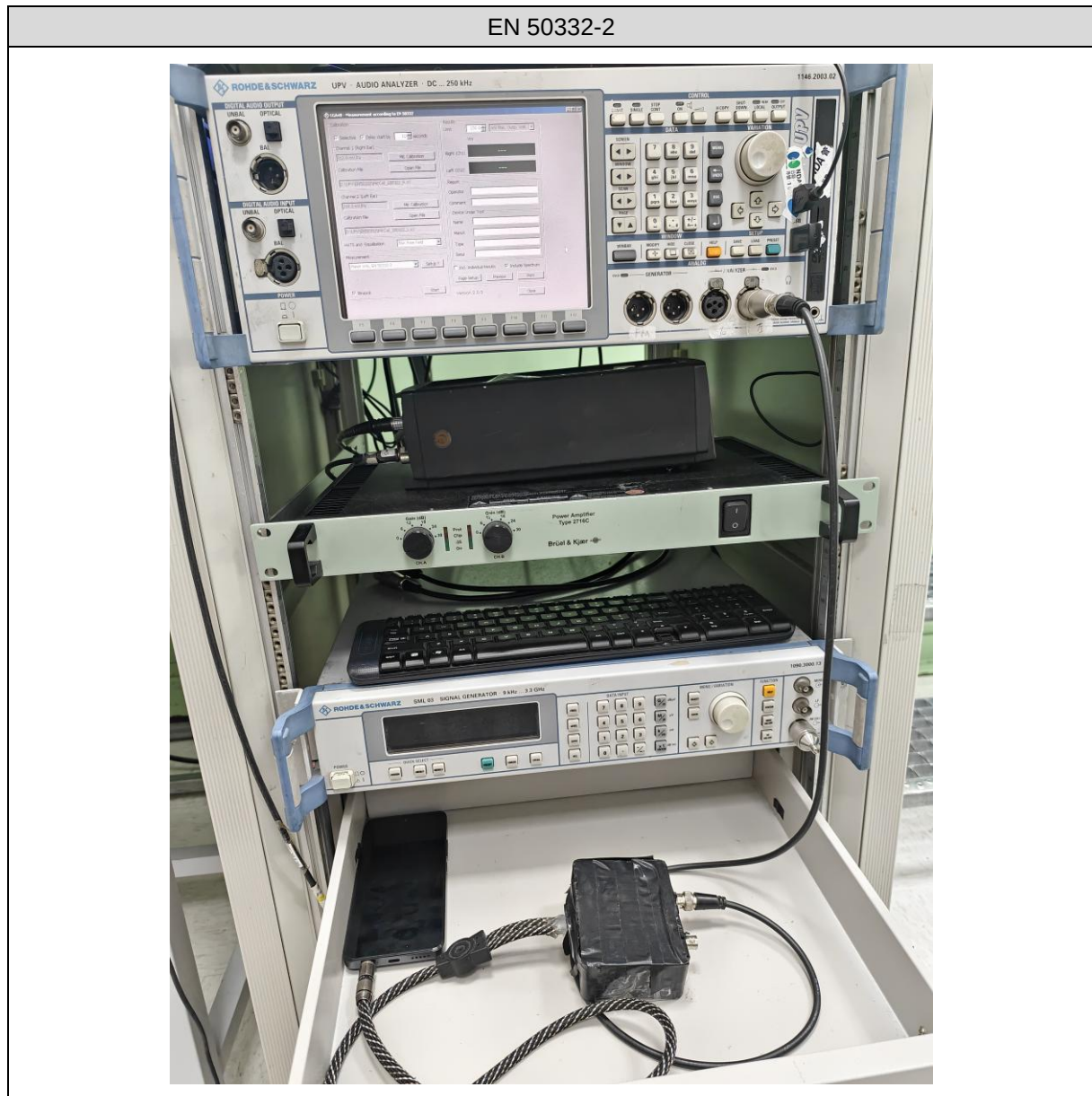
EN50332-2 Analogue Output (mV)			
Left Ear	Right Ear	Limit for RS2	Verdict
92.62	92.56	≤ 150	PASS

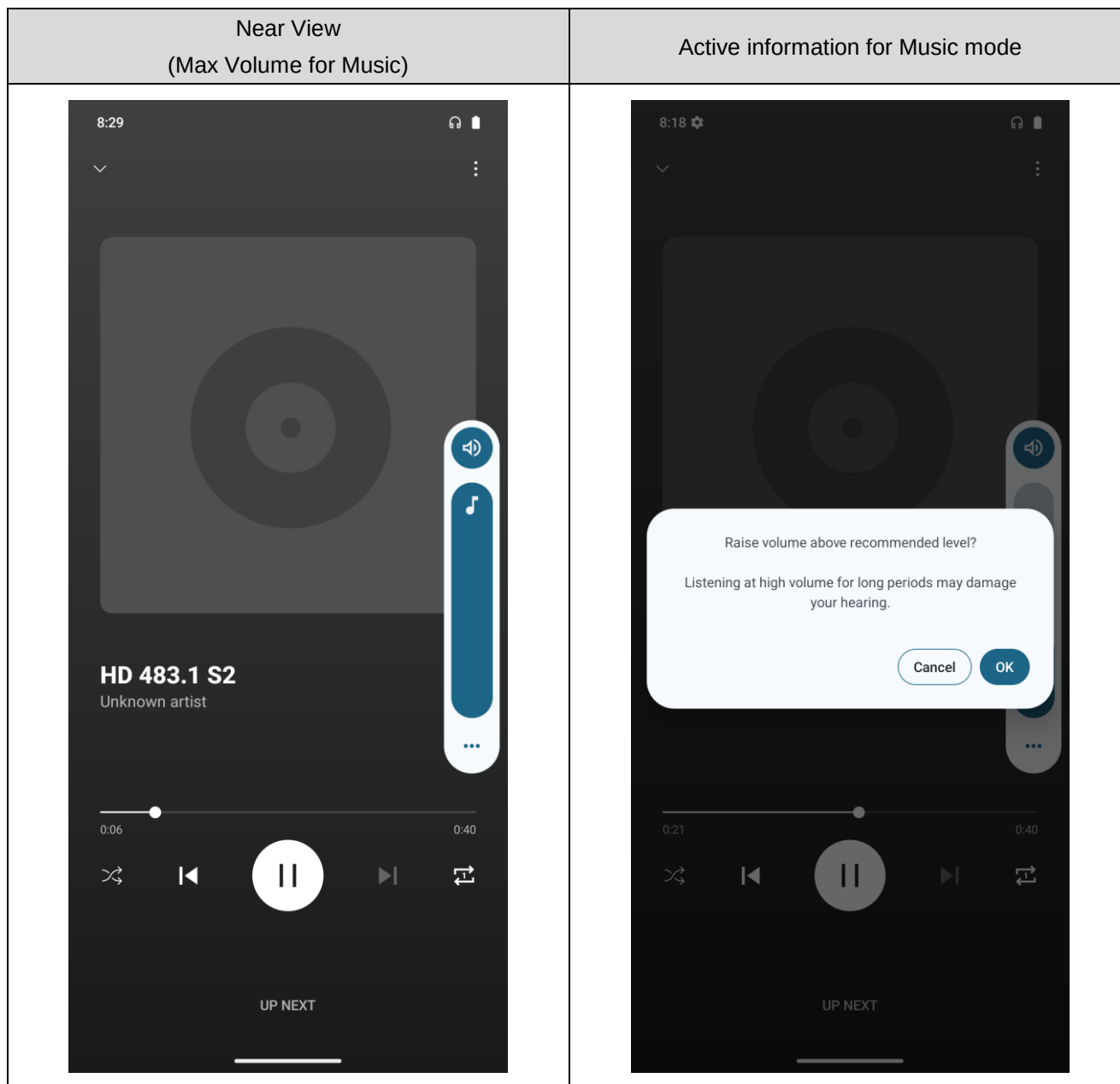
EN50332-2 Type-c Digital output (dBFS)			
Left Ear	Right Ear	Limit for RS2	Verdict
-10.005	-10.010	≤ -10	PASS

4.3 Verify Data When Appearing Warning Information

Player_Music			
For RS2 devices, The output level shall not be higher than RS1 until the acknowledgment is made. Rise volume until the warning text "To prevent possible hearing damage, do not listen at high volume levels for long periods" show up.			
Analogue Output			
Left ear (mV)	Right ear (mV)	Limit	Verdict
3.48	3.44	$\leq 27\text{mV}$	Pass
Output Type-c Digital output (dBFS)			
Left ear	Right ear	Limit	Verdict
-47.704	-47.710	≤ -25	Pass

5. Test Setup Photographs





Front View



Rear View





6. Measurement Uncertainty

Test Item	Uncertainty
Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$) - EN 50332-2 Earphone	0.92mV
Measuring Uncertainty for a Level of Confidence of 95% ($U=2U_c(y)$) - EN 50332-2 Player	0.26mV

————THE END————